

**PROJECT REPORT OF CO-OP PROJECT AT INDUSTRY (MODULE-1)
ON**

Simplifying ESG Evaluation for Small and Medium Enterprises
Submitted in partial fulfilment of the requirements for the award of the degree of

**BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING**

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DECLARATION

We hereby declare that the project work titled, "Simplifying ESG Evaluation for Small and Medium Enterprises. Submitted as part of a Bachelor's degree in Computer Science and Engineering at Chitkara University, Punjab, is an authentic record of our own work carried out under the guidance of the Department of Computer Science and Engineering.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our project supervisor for their continuous guidance, encouragement, and invaluable suggestions throughout the development of this research. Their technical insights and support helped us understand real-world ESG challenges and the practical application of machine learning in sustainability assessment.

We also extend our thanks to the Department of Computer Science and Engineering, Chitkara University, for providing the necessary academic infrastructure, research resources, and a stimulating learning environment that made this work possible.

Finally, we are grateful to our families and peers for their constant encouragement and moral support throughout this journey.

ABSTRACT

Environmental, Social, and Governance (ESG) frameworks have emerged as vital tools for measuring organizational sustainability. However, existing ESG assessment models are predominantly designed for large multinational corporations (MNCs), leaving Small and Medium Enterprises (SMEs) at a systemic disadvantage due to limited resources, unstructured data, and unfamiliarity with ESG practices. This paper proposes a simplified, data-driven ESG evaluation model tailored to the needs of SMEs. Using Exploratory Data Analysis (EDA), threshold-based scoring, and Machine Learning algorithms — including Decision Trees, Random Forest, and Linear Regression — the study classifies organizations into “Safe” and “Risk” categories. The Random Forest classifier achieved 97% accuracy on the test dataset, demonstrating the model’s effectiveness and reliability. The research also performs ESG gap analysis to quantify performance shortfalls against established benchmarks, offering actionable insights for SMEs seeking to improve their sustainability standing.

Keywords: *ESG, Machine Learning, Small and Medium Enterprises, Sustainability, Random Forest, Gap Analysis, Exploratory Data Analysis.*

1. Introduction

1.1. Background

Environmental, Social, and Governance (ESG) criteria have become a cornerstone of modern sustainability assessment. Originally developed to evaluate large corporations, ESG frameworks measure how an organization manages its responsibilities across three interconnected dimensions.

- Environmental: carbon emissions, energy consumption, waste management, water usage, and overall ecological impact.
- Social: labour practices, employee welfare, diversity and inclusion, human rights compliance, and child labour prevention.
- Governance: board transparency, corporate ethics, regulatory compliance, policy integrity, and anti-corruption measures.

1.2. The SME Context

Despite their growing global importance, ESG frameworks suffer from a critical structural bias. Most evaluation tools including widely used Red-Amber-Green (RAG) rating systems are engineered for large MNCs that possess dedicated sustainability teams, structured data systems, and significant financial resources. SMEs, which form the backbone of most economies worldwide, find themselves either excluded from or misrepresented by these frameworks.

A score-based threshold approach helps simplify interpretation. For instance, an ESG score within the range of 0 to 5 may be considered safe, while a score above 5 signals potential environmental or social risk. Such a straightforward framework is far more accessible to SME decision-makers who lack dedicated sustainability expertise.

1.3. Proposed Solution

This paper addresses that gap by introducing a streamlined, machine learning-powered ESG assessment model designed specifically for SMEs. The model uses threshold-based classification, exploratory data analysis, and ensemble learning to deliver accurate, interpretable, and scalable sustainability evaluations. The study draws on globally sourced ESG datasets and evaluates model performance using accuracy, precision, and F1-score as standard metrics.

2. Problem Statement

The rapid growth of ESG-focused investing and regulation has amplified the need for reliable sustainability assessment tools. Yet a fundamental inequity persists: current ESG frameworks are built around the data infrastructure, financial capacity, and institutional knowledge that only large organizations typically possess.

SMEs face a distinct set of structural challenges that make these frameworks inapplicable or misleading:

- Absence of standardized ESG data collection practices.
- Limited financial resources to invest in sustainability auditing and reporting.
- Inadequate awareness and technical know-how regarding ESG metrics and scoring systems.
- Informal labour structures, partial environmental reporting, and incomplete governance documentation.

Consequently, SMEs are either systematically underrated on ESG scales or excluded from sustainability-driven investment opportunities altogether. There is also no established quantitative method to measure the size of this performance gap, making it difficult to prescribe meaningful corrective action.

This research addresses these issues by proposing a simplified, machine learning-based ESG model that:

1. Works within the data constraints typical of SMEs.
 2. Accurately identifies specific weaknesses in ESG performance.
 3. Produces measurable, interpretable outputs using standard machine learning evaluation metrics.
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3. Research Objectives

The primary objective of this research is to develop a data-driven framework for ESG performance assessment in SMEs using machine learning techniques. The specific aims are as follows:

4. To identify and examine the ESG parameters most relevant to SMEs across environmental, social, and governance dimensions.
 5. To collect and preprocess global ESG data to ensure analytical quality and consistency.
 6. To perform Exploratory Data Analysis (EDA) to uncover patterns, trends, correlations, and anomalies within the ESG dataset.
 7. To calculate a composite ESG score from the three core pillars using a weighted combination approach.
 8. To build a machine learning classification model that categorizes organizations as “Safe” or “Risk” based on their ESG scores.
 9. To evaluate model performance using accuracy, precision, and F1-score as standard evaluation metrics.
 10. To conduct ESG gap analysis by comparing actual SME performance against established sustainability benchmarks.
 11. To generate actionable recommendations that help SMEs improve their ESG standing and sustainability reporting practices.
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4. Literature Review

Research on ESG has expanded considerably over the past decade, spanning financial performance, sustainability reporting, and more recently, machine learning applications.

A landmark meta-analysis by Friede, Busch, and Bassen (2015) reviewed over 2,000 empirical studies and found that nearly 90% reported a non-negative relationship between ESG criteria and corporate financial performance, with a substantial majority showing positive outcomes [1]. This established ESG not merely as an ethical obligation but as a driver of long-term value creation.

Johnson and Schaltegger (2016) reviewed two decades of sustainability management tools and noted that SMEs consistently lack the structured processes required for meaningful ESG engagement [3]. SMEs do not typically have organized tools, resources, or standardized approaches to effectively measure and improve their ESG performance, creating a significant divergence between large corporations and smaller enterprises in sustainability assessment.

Berg, Kölbel, and Rigobon (2022) described what they termed “aggregate confusion” the significant divergence in ESG ratings across different rating agencies and methodologies [5]. This inconsistency makes standardized assessment even harder for resource-constrained organizations such as SMEs and underscores the need for more flexible and transparent evaluation frameworks.

On the technological front, Vinuesa et al. (2020) demonstrated that artificial intelligence holds substantial potential in advancing the UN Sustainable Development Goals [6]. Breiman’s foundational work on Random Forests (2001) established the ensemble methodology that underpins the classification model used in this study, offering robustness and high accuracy in complex classification tasks [7].

Taken together, the existing literature confirms both the importance of ESG for SMEs and the notable absence of purpose-built, data-efficient evaluation tools for that segment. This study contributes a machine learning-based solution that directly addresses this identified gap

by integrating EDA, threshold-based scoring, and ensemble classification into a unified SME-friendly framework.

5. Methodology

The research follows a structured pipeline from data collection through model evaluation, designed to produce reliable ESG assessments within the constraints typical of SME data environments. The methodology is illustrated through flowcharts and step-by-step descriptions below.

5.1. Data Collection

The dataset is assembled from globally sourced, publicly available ESG repositories to ensure diversity and reliability. These include:

- Open sustainability datasets.
- Public ESG reports from international organizations.
- Financial and environmental indicators from sources such as the World Bank [8].

The dataset is structured across three primary dimensions:

- Environmental Data: factors including carbon emissions and energy consumption.
- Social Data: indicators covering labour practices and employee welfare scores.
- Governance Data: metrics on transparency, board structure, and compliance.

5.2. Data Preprocessing

Prior to model development, all data undergoes a four-step preprocessing pipeline to ensure consistency and analytical readiness:

12. Missing values are handled using statistically appropriate imputation techniques.
13. Duplicate records are identified and removed to eliminate redundancy.
14. ESG scores are normalized to a uniform scale for cross-comparability across organizations.
15. Categorical variables are encoded into numerical format for compatibility with machine learning algorithms.

5.3. Exploratory Data Analysis (EDA)

EDA is conducted to understand the distributional properties of the dataset and guide model selection. Key analyses include:

- Density and histogram plots across ESG pillars to understand score distributions and identify skewness (Figs. 1 and 2).
- Q-Q plots confirming that total ESG scores deviate from normality, particularly at the tails, indicating non-Gaussian behaviour (Fig. 3).
- Pearson correlation analysis revealing that the environment score ($r = 0.959$) is the strongest predictor of total ESG score, followed by social ($r = 0.810$) and governance ($r = 0.712$) dimensions (Fig. 4).
- Box plots showing consistent ESG score variation across 12 industry categories, confirming that large MNCs systematically outscore SMEs across all dimensions (Figs. 6 and 7).

5.4. Threshold-Based ESG Evaluation

To simplify interpretation, a threshold-based classification system is applied to ESG scores:

- Safe Zone: ESG scores within the range of 0 to 5 — the organization is functioning within acceptable sustainability parameters.
- Risk Zone: ESG scores above the threshold of 5 — indicating potential environmental, social, or governance harm.

This binary framing simplifies ESG interpretation significantly for SME decision-makers who may not have the expertise to navigate continuous, complex scoring systems.

5.5. Machine Learning Model Development

Three machine learning algorithms are employed to analyze ESG data and predict performance:

- Linear Regression: applied for continuous ESG score estimation and as an interpretable baseline.
- Decision Tree: used for rule-based classification of organizations into ESG performance categories.
- Random Forest: an ensemble method that enhances classification accuracy and model robustness through the aggregation of multiple decision trees [7].

All models are trained on a split dataset (training and testing subsets) and evaluated for generalization performance on unseen data.

5.6. Model Evaluation Metrics

Model performance is assessed using three standard metrics:

- Accuracy: the overall proportion of correctly classified instances.
- Precision: the reliability of positive (Risk) predictions, minimizing false alarms.
- F1-Score: the harmonic mean of precision and recall, providing a balanced evaluation relevant for datasets with class imbalance.

5.7. ESG Gap Analysis

The ESG gap is quantified by comparing each organization's actual ESG score against a predefined benchmark value. This analysis identifies the magnitude of underperformance and highlights specific pillars where SMEs most frequently fall short. KDE (Kernel Density Estimation) plots confirm that large MNCs cluster around positive ESG gaps, while SMEs skew consistently negative across all dimensions (Fig. 8).

6. Implementation Details

6.1. Frameworks and Tools

The framework is implemented in Python, chosen for its rich ecosystem of data science and machine learning libraries. The following tools are used:

- NumPy: numerical computation and array operations.
- Pandas: data manipulation, cleaning, and preprocessing.
- Matplotlib and Seaborn: data visualization throughout the EDA phase.
- Scikit-learn: model training, cross-validation, and evaluation.
- Jupyter Notebook and VS Code: interactive development and experimentation environments.

6.2. Dataset Implementation

The dataset includes ESG-related indicators classified into the three broad dimensions described in Section 5.1. All categorical values are converted into numerical format prior to model training. Carbon emission figures and energy usage represent environmental features; employee welfare and labour compliance scores represent social features; transparency scores and board structure metrics represent governance features.

6.3. Core Logic

The system implements a sequential pipeline of eight steps:

16. Data loading and initial inspection of structure and data types.

17. Preprocessing: handling missing values, removing duplicates, and selecting meaningful ESG features.
18. ESG score computation via a weighted combination of environmental, social, and governance sub-scores to ensure each dimension is proportionally represented.
19. Threshold-based classification of organizations into Safe or Risk categories based on their composite ESG score.
20. Exploratory Data Analysis via histograms, density plots, correlation matrices, and Q-Q plots.
21. Dataset split into training and testing subsets; Random Forest model training on selected ESG features.
22. Model evaluation using accuracy, precision, and F1-score on the held-out test set.
23. ESG gap computation by comparing each organization's ESG score to a predetermined benchmark, exposing deviations in performance.

6.4. System Output

The implemented system produces four primary outputs for each organization:

- A composite ESG score representing overall sustainability performance.
- A binary classification label (Safe or Risk) based on the threshold system.
- A gap value indicating the distance between the organization's score and the benchmark.
- A model performance report including accuracy, precision, and F1-score.

6.5. Integration and Scalability

The framework is designed with extensibility in mind. It can be integrated into a full-stack web application using a Flask-based Python backend that exposes the trained model as a REST API. A React-based frontend dashboard would enable SMEs to input their own ESG indicator values and receive real-time scores, classifications, and gap analyses without requiring technical expertise.

7. Major Findings / Results

7.1. EDA Findings

The EDA reveals several important structural patterns in global ESG data. Score distributions across the three pillars are non-normal, with heavy tails particularly in the governance dimension. Pearson correlation analysis confirms that environmental performance is the dominant driver of total ESG scores ($r = 0.959$), while governance, though important, has a comparatively weaker correlation ($r = 0.712$). K-Means clustering identified two optimal clusters ($k = 2$) based on silhouette scoring, aligning closely with the SME versus MNC segmentation present in the data.

7.2. ESG Gap Analysis

The gap analysis is the most policy-relevant finding of this study. KDE plots clearly demonstrate that the ESG gap distribution for large MNCs is centred in positive territory, while the distribution for SMEs skews negative across all three pillars (Fig. 8). This is not merely a data artifact; it reflects structural disadvantages — SMEs typically operate with informal governance practices, limited environmental reporting infrastructure, and weaker labour compliance documentation.

Box plots further confirm that large MNCs consistently outperform small and other companies across all ESG dimensions (Fig. 9), reinforcing the systemic nature of the assessment gap.

7.3. Model Performance

The Random Forest classifier achieved 97% overall accuracy on the test dataset. The detailed classification report is as follows:

Category	Precision	Recall	F1-Score
Risk	0.96	0.99	0.97
Safe	0.98	0.96	0.97
Overall Accuracy	0.97	—	0.97

Feature importance analysis from the Random Forest model shows governance score as the most influential predictor, followed by social and environment scores. This suggests that governance reforms may deliver the highest ESG impact per unit of effort for SMEs.

8. Conclusion and Future Scope

8.1. Conclusion

This study presents a practical, machine learning-powered framework for ESG assessment tailored to the realities of Small and Medium Enterprises. By combining threshold-based scoring, exploratory data analysis, and Random Forest classification, the model delivers a scalable and interpretable solution that bridges the gap between existing ESG tools designed primarily for MNCs and the operational realities of SMEs.

The Random Forest classifier's 97% accuracy on ESG classification, paired with a quantitative gap analysis methodology, demonstrates that sophisticated sustainability assessment is achievable without large-scale data infrastructure. The framework not only simplifies ESG evaluation but also produces actionable outputs that SMEs can use to prioritize improvements across environmental, social, and governance dimensions.

8.2. Future Scope

- **Real-time ESG Data Feeds:** integrating live sustainability data streams to enable continuous monitoring rather than periodic reporting.
 - **Sector-Specific Benchmarks:** extending the model to industry-specific ESG benchmarks for more targeted gap analysis.
 - **SME Dashboard:** developing a user-facing web application that allows SMEs to self-assess and track their ESG progress over time.
 - **Broader Dataset Coverage:** expanding the dataset to include more developing-economy firms to strengthen generalizability across diverse regulatory and economic contexts.
 - **Advanced ML Techniques:** exploring deep learning and NLP-based ESG signal extraction from unstructured reports and news sources.
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9. REFERENCES

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